Part 2: Revised Proposal

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Project Title: AI-Gamified Discord Bot for Enhancing Collaboration in Remote GitHub Teams

Domain of Interest

We focus on communication challenges in remote work settings, specifically within distributed software development teams using GitHub. This domain aligns with AI's role in information technology organizations. It will examine how AI can mitigate issues in asynchronous collaboration, such as fragmented discussions, reduced team engagement, and motivation. These are common in global IT firms and open-source projects, where text-based workflows often lead to passive participation. For instance, lurkers who observe but rarely contribute.

The Problem We Are Trying to Solve and Why

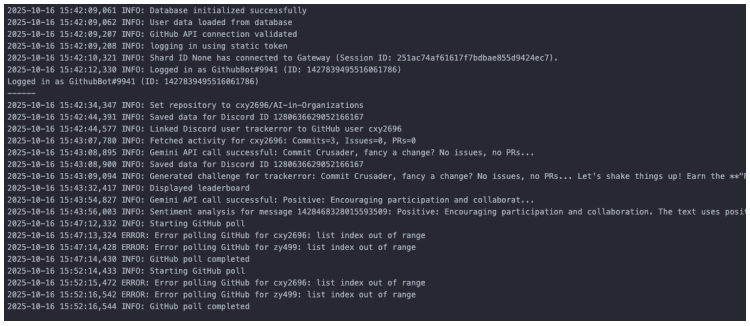
GitHub projects rely heavily on text-based and asynchronous workflows, which can result in lost follow-ups in threads and low motivation for tasks like code reviews. THis contributes to decreased team cohesion and productivity in remote settings. We are building an AI tool to introduce gamification, making routine interactions rewarding and fostering engagement in the GitHub community through a Discord bot. This provides a practical AI solution to improve remote team dynamics in IT organizations, addressing gaps in motivation and collaboration that traditional tools overlook.

Target Users

Developers or contributors in distributed GitHub teams using Discord for communications, including lurker contributors who read issues and pull requests (PRs) but seldom commit. Team leaders who monitor project progress and seek ways to boost members' participation.

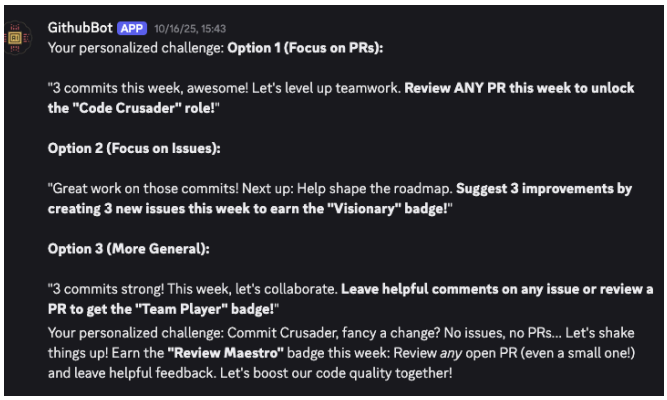
Updated Project Plans

We have prototyped an AI tool as a Discord bot (named “GithubBot” so far) that integrates with GitHub API and Gemini 2.0 API to gamify interactions. Initially using a Python-based implementation for control over stateful features and API integrations. We used some standard libraries (e.g. Github, Auth in github, commands in discord.ext, and urllib for Gemini API calls) to build a robust prototype). The record of the test for comments in Discord channel is shown below.

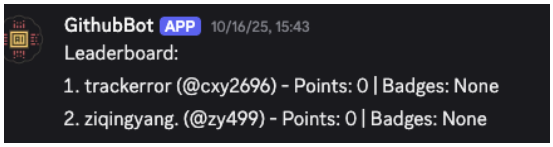


Key AI features implemented:

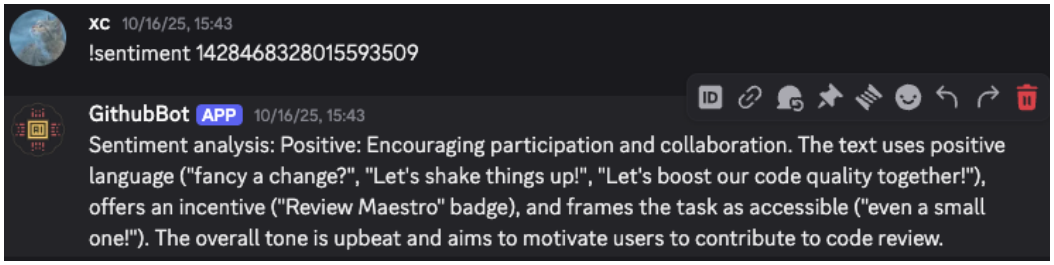
- AI-generated personalized challenges (through a customized command “!my_challenge): Using Google’s Gemini API, the bot analyze user GitHub activity (commits, issues, PRs) since the last check and generates tailored challenges (e.g. {Review on PR to earn the “Review master” badge}). This encourages low-activity users to engage.



- Progress tracking with leaderboards, points, and badges (through a customized command "!leaderboard): The bot will poll GitHub every 5 minutes (current setting), awards points (calculation function in progress), and then update badges to display leaderboards.



- Sentiment analysis (through customized comment "!sentiment {message_id}): Gemini analyzes Discord message sentiment for feedback to help team members evaluate discussion health.



Needed resources (get): GitHub API for data pulls (issues, pull requests, commits). Discord API for bot integration and real-time interactions. Gemini API for challenge generation and sentiment analysis.

Revised Goals

Core Goal: Deliver a functional prototype that demonstrates gamification's impact on collaboration, with an analysis of its organizational effects (e.g., tracking PRs) and societal implications (e.g., potential biases in AI personalized challenges or sentiment analysis). New Insights: Feedback from internal testing revealed error handling needs (GitHub polling errors like "list index out of range" for linked users, and the point calculator function refinement). This is leading to refined logging and validation. Clarified the channel to focus on Discord over Slack

for easier setup and reducing integration complexity. Ethical considerations will emphasize bias mitigation in challenges.

Updated Timeline for the Rest of the Semester

- Week 9 (10/27-11/2, completed): Internal testing of basic functionality (e.g., linking users, generating challenges, polling activity) and mid-semester check-in (demo on 10/16).
- Week 10 (11/3-11/9): Refine prototype, adding robust error handling, expand sentiment analysis to flag negative discussions, and conduct small-scale user testing with 3-5 peers simulating a GitHub repo.
- Week 11-12 (11/10-11/23): Ethical auditing (e.g., bias checks in AI outputs) and iterate based on test feedback; prepare for organizational impact analysis (e.g., points set on engagement).
- Week 13 (11/24-11/30): Finalize showcase materials, including video demo and report sections; simulate real-world adoption scenarios.
- Week 14/15: Final showcase and submission.

Part 3: Drafted Abstract and Methodology

Abstract

In the evolving landscape of remote work, distributed software development teams on platforms like GitHub often face challenges in maintaining engagement and collaboration. That's because of issues like asynchronous, text-heavy workflows. This project introduces an AI-gamified communication channel, a Discord bot integrated with GitHub API, which helps to enhance team dynamics by incentivizing participation through personalized challenges, points,

badges, and sentiment analysis. Motivated by common issues such as low motivation for code reviews and fragmented discussions, our tool uses Google's Gemini API to generate tailored challenges based on user activity (e.g., "Review a PR to earn a badge") and analyze message sentiment for constructive feedback.

Our approach involves prototyping a Python-based bot that polls GitHub for contributions, awards points (e.g., 10 for commits), updates leaderboards, and stores data securely in SQLite. Preliminary internal testing on a test repository demonstrated functionality: users linked accounts, received challenges, and displayed the leaderboard, with sentiment analysis identifying positive tones in motivational messages.

The key contribution is a practical AI solution that bridges gamification with remote collaboration, potentially boosting productivity in IT organizations. By addressing biases in personalization (e.g., inclusive prompts for low-activity users), the tool offers insights into AI's role in fostering equitable team environments. Future iterations will include user testing to quantify impacts, highlighting societal implications like reduced isolation in global teams and ethical considerations in AI-driven motivation systems.

Methodology

Our methodology focuses on prototyping an AI tool to gamify GitHub-Discord interactions, emphasizing design, implementation, and evaluation to address remote collaboration challenges. We adopted an iterative agile approach, starting with requirements gathering from personal experiences and literature on gamification in software teams (e.g., studies on points/badges increasing contributions).

Design and Process: The tool's core is a Discord bot built in Python 3.12, using discord.py for bot commands, PyGithub for API access, and urllib for Gemini API calls. Key

processes include: User Linking and Activity Polling: Users link Discord to GitHub via "!link_github" command. The bot polls the repository every 5 minutes to fetch commits, issue comments, and PR reviews since the last check, awarding points accordingly (e.g., 10 points/commit, 15/PR review). Data is stored in an SQLite database for persistence, with timestamps in UTC to handle global users. AI Personalization: Gemini API generates challenges with prompts like "Based on activity: Commits=3, Issues=0, PRs=0. Generate an engaging challenge." This addresses low engagement by suggesting actionable tasks. Sentiment analysis uses similar prompts on message content for comment summaries. Gamification Elements: Points trigger badges (e.g., "Bronze Collaborator" at 10 points); leaderboards are displayed via "!leaderboard". Challenges offer bonus points upon completion.

Tools and Frameworks: APIs: GitHub for event data; Gemini for NLP tasks (rate-limited with retries); Discord channel bot for real-time interaction. Evaluation Criteria: Functionality tested via internal simulations (e.g., fake commits in a test repo like "cxy2696/AI-in-Organizations"). Metrics include points earned, challenge completion rates, and error logs. Qualitative: Sentiment outputs and user feedback on motivation. Ethical framework: Bias checks ensure challenges are positive and inclusive; no sensitive data (e.g., geolocation) is used.

This methodology ensures the prototype is scalable and testable, with room for refinement based on feedback. Limitations include API dependencies. The future work will expand to quantitative surveys measuring engagement uplift.